

# **realpath / readlink**

Relative paths and symlinks are handy, until a script needs one clear absolute location

# Stored target vs canonical path

- Readlink on a symlink prints the stored target string, often a short relative path
- Realpath (and readlink with a force-style flag on Linux) cleans dots and parent segments
- That difference matters in build scripts

# Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / realpath

INTERACTIVE TOOL

realpath / readlink

Contrast readlink (stored target string) with realpath / readlink -f (canonical path after following links).

Starter example: cwd is /home/lab/chip/links. Compare readlink to\_top (prints ../rtl/top.v) with realpath to\_top (canonical /home/lab/chip/rtl/top.v). Then try chain\_a and the broken gone.

Load starter example

Challenges 0 / 22 cleared

Quiz: readlink: readlink prints the? Answer: target or string

Answer

Show hint

Check

Next

Idle

Quiz: readlink

Quiz: realpath

readlink to\_top

realpath to\_top

Contrast answer

Canonical answer

Chain realpath

Chain readlink

Broken readlink

Broken realpath

Absolute link

Deep relative

Quiz: relative base

cd through link

Sibling from rtl

readlink -f

Toolchain link

Quiz: not the same

Dir symlink

Compare panel

Plain file

Quiz: broken

Lab terminal

pwd /home/lab/chip/links

Starter loaded - try: readlink to\_top + realpath

Side-by-side resolve

Path

to\_top

Compare

realpath resolve lab

# Real shell practice

```
wsl - module06-realpath-resolve/examples/realpath

# Track A - real Unix
# real shell session (Track A)

$ pwd
/mnt/d/proj/designs/learning/courses/learn_unix/module06-realpath-resolve/examples/realpath

$ realpath .
/mnt/d/proj/designs/learning/courses/learn_unix/module06-realpath-resolve/examples/realpath

$ realpath a_file.txt
/mnt/d/proj/designs/learning/courses/learn_unix/module06-realpath-resolve/examples/realpath/a_file.txt

$ ls -l link_to_file
lrwxrwxrwx 1 yongfu yongfu 10 Jul 20 03:00 link_to_file -> a_file.txt

$ readlink link_to_file
a_file.txt

$ realpath link_to_file
/mnt/d/proj/designs/learning/courses/learn_unix/module06-realpath-resolve/examples/realpath/a_file.txt

$ readlink -f link_to_file
/mnt/d/proj/designs/learning/courses/learn_unix/module06-realpath-resolve/examples/realpath/a_file.txt
```

Real shell — realpath and readlink

## Real shell practice — try these

```
# pwd — where you are before resolving  
pwd
```

```
# realpath . — absolute canonical path of the current directory  
realpath .
```

```
# realpath a_file.txt — absolute path of the sample file  
realpath a_file.txt
```

```
# ls -l link_to_file — see the symlink arrow to the file  
ls -l link_to_file
```

```
# readlink link_to_file — stored target string (often relative)  
readlink link_to_file
```

```
# realpath link_to_file — follow the link to a canonical absolute path
```

# Pitfalls to watch

- A broken link still has a stored string
- Do not assume every system has the same readlink flags
- And remember

# Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On the real shell, compare readlink's stored target with realpath's canonical path
- When you are ready, take the short quiz, then continue to permissions, umask, and PATH